

Responsible chair:	Control & Simulation
Principal Tutor:	dr. ir. M. M. van Paassen
Project title:	AHAPS; Adaptable High Altitude Platform System
Mission Objective	Create an accessible and low-cost modular high altitude platform system (aka high altitude pseudo satellite), that can be easily transported to its region of deployment, assembled with limited tooling, adapted to different missions, providing a four week or more endurance.
Keywords	UAV, link budget, autonomous flight, solar power, endurance, battery technology, high altitude flight, modular platform.
Abstract	A High Altitude Platform Station is like a poor nation's satellite, offering the capabilities of a low-orbit satellite, but at a more local scale, for example radar observation for crop monitoring and pest control, thermal imaging supporting forest fire detection, or high-bandwidth communications needed locally in remote regions. In this project, the Adaptable High Altitude Platform System is designed. The AHAPS is an adaptable HAPS, low-cost, light weight and transportable in a modestly sized trailer. It can be assembled and launched in the field with limited tooling, and offers services for different types of payload. It can be applied in disaster response, at times of forest fire risk, for crop and nature monitoring, or as a research platform for universities and research institutes, and can service an area of approximately 400x400 km.

Introduction Satellite-based earth observation, communication and localization provide many useful services. However, by their very nature these are globally functioning devices, lead times and cost for launch are large, these platforms are primarily operated by large and developed industrial nations. As an alternative, high-altitude autonomous vehicle technology can provide a more flexible platform for local needs. Such a High Altitude Platform Station offers the possibility of a low-orbit satellite, but at a more local scale, for example radar observation for crop monitoring and pest control, thermal imaging supporting forest fire detection, or high-bandwidth communications needed locally in remote regions. Several HAPS have been designed and tested, with most of these being prototypes. However, with current advances in light-weight materials, batteries, computation and electric motors, it should be possible to create a new generation of HAPS that can take up a niche roles in local communications, disaster response, crop and pest monitoring. The design assignment is for a flexible platform, that would support various payloads, such as (1) communications equipment, (2) synthetic aperture radar, (3) visual and infra-red range cameras, (4) various scientific payloads, e.g. for university/research institute use, with an emphasis on accessibility (low-cost, transportable by trailer, and deployable from locations with limited ground facilities).

Requirements Specify the requirements and constraints precisely, i.e., unambiguously, quantitatively and verifiably:

- Performance:
 - Minimum cruise altitude: 60000 ft
 - Minimum endurance: 28 days, all-season, between -60SB and 60NB
 - Autonomous loitering operation
 - Take-off and landing from unpaved but prepared flat surface, possibly vehicle-assisted
 - 400 km horizontal communication range
- Safety and reliability:
 - The aircraft only needs intermittent (daily) monitoring
 - A health monitoring system supports operational decision making
 - equipment includes at least an SSR transponder, ADSB transmitter, ELT beacon
 - EASA Civil drones certification, operations type #1 applies
- Sustainability:
 - Materials are recyclable or inert when disposed of
 - Batteries should have a service life of at least 2 years, at 50% (180 day) yearly use
- Engineering budgets:
 - Payload mass: 20 kg
 - Payload energy use: intermittent (5 min window) peaks of 250 W, average 100W.
 - Payload heat dissipation: 150W.
 - Possibility to mount a 1x1 m synthetic aperture radar antenna (weight included in payload weight)
 - Platform provides a communications link providing control, system monitoring, and 1Mbit/s data for the payload
 - Concept engineering (DSE phase) 7 weeks, 10 persons
 - TBD
- Cost:
 - Materials and manufacturing cost per unit: 300,000 euro.
- Other:
 - Disassembled transport in custom trailer with 4x2 m loading space
 - AHAPS with trailer is transportable in standard 20 feet ISO container
 - Final assembly in the field with limited tooling

Review Milestones, Objectives and Deliverable Items Kickoff

- Beginning P2/P4 - see schedule on Brightspace.
- Objective: to start the project and to get acquainted

2. Project Plan (PP)

- Date: End of the 1st week of the DSE
- Objectives: Consolidating a provisional, systematic planning and project procedures
- Deliverable items:
 - i. Project plan Report containing the items in the table below.

3. Baseline Review (BR)

- Date: see schedule
- Objectives: Review the requirements, the planning and initial concepts
- Deliverable items:
 - i. Baseline report containing the items in the table below.
 - ii. Reports have to be handed in at least 24 hours before the Baseline Review (date can be preponed by the tutor)

4. Mid-Term Review (MTR)

- Date: see schedule
- Objectives: Review of the concepts and trade-off process of the final concepts
- Deliverable Items:
 - i. Mid-Term Report containing the items in the table below.
 - ii. Report have to be handed in handed in at least 24 hours before the Midterm Review (date can be preponed by the tutor)

5. Final Review (FR)

- Date: see schedule
- Objectives: Technical presentation and review of the final design
- Deliverable Items:
 - i. Final Report containing the items in the table below
 - ii. Report has to be handed in handed in at least 24 hours before the Final Review (date can be preponed by the tutor)

6. Symposium Day

- Date: see schedule
- Objectives: Enjoy yourself and inform friends & family
- Deliverable Items:
 - i. Presentation
 - ii. Poster
 - iii. Executive Summary
 - iv. Jury summary

<i>No</i>	<i>Product</i>	<i>Project Plan</i>	<i>Baseline Report</i>	<i>Mid-term Report</i>	<i>Final Report</i>
1	DSE work-flow diagrams	•		•	
2	DSE work break-down structure	•		•	
3	DSE Gantt chart	•		•	
4	DSE team organization/organogram & HR allocation	•		•	
5	Functional flow diagram(s)		•		•
6	Functional breakdown		•		•
7	Requirements discovery tree		•		
8	Resource allocation/ Budget breakdown		•		•
9	Technical risk assessment/risk map		•	•	•
10	Design option structuring (tree)		•	•	
11	Contingency Management		•		
12	Market analysis		•		•
13	Interface definition/N2 charts			•	
14	Trade method, rationale and organization			•	
15	Trade criteria			•	
16	Criteria weight factors			•	
17	Trade summary table			•	
18	Operations and logistic concept description			√	√

No	Product	Project Plan	Baseline Report	Mid-term Report	Final Report
19	Project design & development logic		√	√	√
20	Project Gantt chart				•
21	Cost break-down structure				•
22	H/W, S/W block diagrams (interactions, flows)				√
23	Electrical block diagram				√
24	Data handling block diagram				√
25	Sustainable development strategy		•	•	•
26	Compliance Matrix (table with tick marks for all requirements that are met)				•
27	Sensitivity Analysis			•	•
28	Verification & Validation Procedures			•	•
29	Communication flow diagram				√
30	Manufacturing, Assembly, Integration plan (production plan)				√
31	Return on investment, operational profit				√
32	Reliability, Availability, Maintainability, and Safety (RAMS) characteristics				√
33	Performance analysis (e.g. flight profile diagrams, payload-range diagrams, climb performance, noise characteristics, emissions, etc.)			√	√
34	Configuration / layout (internal /external)			√	√
35	Spacecraft system characteristics (e.g. communications link budget, memory size, etc.)				
36	Aircraft system characteristics (e.g. fuel and hydraulic system lay-out, auxiliary power estimate, environmental control, etc.)			*	*
37	Aerodynamic characteristics estimate (e.g. lift, drag, aerodynamic moments, drag polar, etc.)				√
38	Structural characteristics (e.g. loading diagrams, stresses, bending, flutter, etc.)				√
39	Stability and control characteristics (e.g. control forces, c.g. limits, etc.)				
40	Material characteristics (e.g. yield, ultimate, fatigue, etc.)				
41	Astrodynamic characteristics (e.g. orbit, trajectory, decay rate, ΔV-budget)				

Note: preset (•), set by PT (√), (*) adhere to item 35 OR 36 as applicable.

Project Outline

Nr	Discipline	Relative importance %
1	Aerospace materials	5
2	Manufacturing	5
3	Structures	10
4	Aerodynamics	10
5	Stability and Control	5
6	Design integration and system Engineering	10

7	Performance	10
8	Propulsion	10
9	Operations	10
10	Wind energy	
11	Astro dynamics	
12	Remote sensing and Planetary Science	5
13	Sustainable engineering	10
14	Other (atmosferic modelling, wind conditions)	10

Internal and external experts

Internal experts

Name	Expertise	Not available in the period
C. Varriale	Flight mechanics	
B. Remes	UAV	

External Experts

Name	Expertise	Not available in the period
P. Lopez Dekker	Remote sensing, radars (CITG)	

References

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